

Effect of dietary sodium restriction on taste responses to sodium chloride: a longitudinal study.

Normotensive adults on low-sodium, weight-loss, and control diets recorded preferences and perceived saltiness for sodium chloride (NaCl) added to cream soup at intervals over 1 yr. Reduction in sodium intake and excretion accompanied a shift in preference toward less salt: preferred concentrations by ad libitum salting declined from 0.72% at the onset to 0.33% NaCl at week 24; hedonic scores for high concentrations of NaCl decreased significantly while scores for low concentrations increased. After 3 mo of sodium restriction, NaCl preferences readjusted to a lower level: ad libitum additions of NaCl were similar after 13, 24, and 52 wk. Less hedonic variation was observed among controls than among Na-restricted groups. The weight-loss group showed increased liking for mid-range NaCl levels. Mechanisms underlying preference changes, including physiological, behavioral, and context effects, may provide insights into maintenance of low-sodium diets for treatment and prevention of hypertension.